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# The Algebra of Narratives and the Monitoring of Their Relative Dominance

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**Abstract**

We propose an algebraic methodology for measuring the relative prevalence of competing narratives within a time-evolving corpus of texts. Standard cosine similarity between article embeddings and narrative vectors fails to isolate the idiosyncratic contribution of each narrative, because narratives sharing a common topic produce strongly correlated vector representations. We address this by orthogonalizing the narrative vectors via a global max-min optimization on the Stiefel manifold, which neutralizes shared semantic components while preserving each narrative's specific direction. We ground this geometric procedure in a linear generative model for article embeddings that provides an unbiased estimator of narrative prevalence and formally explains why the naive approach suffers from a correlation bias. Empirical validation on the publicly available News Category Dataset (HuffPost, 2012–2018) confirms that the orthogonalized estimator recovers narrative-specific dynamics aligned with known historical events, whereas the naive cosine projection yields synchronised, uninformative time series. **[TODO: Refine abstract once results section is complete.]**

**Keywords:** narrative analysis, Stiefel manifold, text embeddings, orthogonalization, semantic similarity

## 1. Introduction

Since the emergence of vector representations of texts, their power has been evidenced by the structure of distance in the Euclidean space to which they belong. Early demonstrations include the native clustering of Self-Organizing Map (SOM) prototypes on Grimm tales (Honkela, Pulkki, and Kohonen 1995) and the celebrated *king* − *man* + *woman* = *queen* relation satisfied by doc2vec embeddings (Mikolov, Yih, and Zweig 2013). Modern bidirectional encoders such as BERT have turned semantic similarity into a practical primitive, notably in Retrieval-Augmented Generation (RAG) pipelines (Wang et al. 2024).

Concurrently, the concept of *narrative competition* has gained traction in economic literature (Shiller 2020): narratives are the mean field through which opinions form, and agents condition their decisions on the perceived relative dominance of narratives. Institutional actors exploit this dynamic; for instance, the Bank of England instruments narratives to guide market expectations (Naef 2024). Market participants continuously probe and reshape narratives in a closed feedback loop that renders straightforward causal analysis difficult.

The narrative viewpoint differs fundamentally from sentiment analysis. Rather than classifying a document as positive or negative—an approach that conflates semantic and frequentist synonyms (Li et al. 2024)—it uses the alignment of a text with the vector representation of a narrative as a measure of *importance*. Formally, the cosine similarity between a news article and a narrative takes values

close to +1 when they are perfectly aligned, close to −1 when contradictory, and close to zero when semantically independent in the high-dimensional embedding space.

**[CAL: todo, biblio w/r to gender neutrality]**

Challenges. Measuring relative narrative strength poses two distinct challenges.

1. **Idiosyncratic vs. common co-movements.** Even apparently opposed narratives—say, "Clinton will win the election" and "Trump will win the election"—share a common semantic direction (the *elections* axis) that causes their cosine scores to rise and fall in unison for reasons unrelated to which candidate is actually gaining ground. Disentangling idiosyncratic from common variation is therefore essential. **[CAL: je trouve délicat de mettre en exemple qu'on ne traite pas dans le papier; est-ce que ça fait sens de les rajouter?]**

Contributions. We propose an algebraic framework that simultaneously addresses both challenges. Our key contributions are:

- A global orthogonalization procedure on the Stiefel manifold that constructs, for each narrative, a representative vector that is close to the original narrative while being orthogonal to all others (Section 5).
- A linear generative model for article embeddings that provides a rigorous statistical justification for the procedure, yielding an unbiased estimator of narrative prevalence and formally exposing the correlation bias of the naive approach (Section 3).
- An empirical validation on a publicly available labeled news dataset, demonstrating qualitative and quantitative agreement with known historical events (Section 6).

## 2. Intro (bis)

Since the emergence of vector representations of texts, their power has been evidenced by the structure of distance in the Euclidean space to which they belong. Early demonstrations include the native clustering of Self-Organizing Map (SOM) prototypes on Grimm tales (Honkela, Pulkki, and Kohonen 1995) and the celebrated *king* − *man* + *woman* = *queen* relation satisfied by doc2vec embeddings (Mikolov, Yih, and Zweig 2013). Modern bidirectional encoders such as BERT have turned semantic similarity into a practical primitive, notably in Retrieval-Augmented Generation (RAG) pipelines (Wang et al. 2024). Concurrently, the concept of *narrative competition* has gained traction in economic literature (Shiller 2020): narratives are the mean field through which opinions form, and agents condition their decisions on the perceived relative dominance of narratives. Institutional actors exploit this dynamic; for instance, the Bank of England instruments narratives to guide market expectations (Naef 2024). Market participants continuously probe and reshape narratives in a closed feedback loop that renders straightforward causal analysis difficult.

The narrative viewpoint differs fundamentally from sentiment analysis. Rather than classifying a document as positive or negative—an approach that conflates semantic and frequentist synonyms (Li et al. 2024)—it uses the alignment of a text with the vector representation of a narrative as a measure of *importance*. Formally, the cosine similarity between a news article and a narrative takes values close to +1 when they are perfectly aligned, close to −1 when contradictory, and close to zero when semantically independent in the high-dimensional embedding space.

T his paper addresses the following question: how can one monitor, over time and at scale, the relative prevalence of competing narratives within a large corpus of text? Existing approaches largely fall into three families. The first relies on keyword or dictionary-based counting: canonical examples include the Economic Policy Uncertainty index of (Baker, Bloom, and Davis 2016),

built from keyword counts in newspaper archives, and the finance-specific sentiment dictionaries of (Loughran and McDonald 2011). Being built on exact vocabulary matches, this family offers limited semantic granularity: a narrative expressed through paraphrase or implication, without the target keywords, goes undetected. The second relies on unsupervised topic modeling, typically via Latent Dirichlet Allocation applied to financial or economic news (**larsen2019business**; Bybee et al. 2020), or, closer to our setting, on text-semantics pipelines that extract narrative statements from syntactic structure (**ash2021relatio**). Although richer than raw keyword counts, this family remains fundamentally a bag-of-words approach: topics are inferred from word co-occurrence statistics across the corpus, with no representation of synonymy or paraphrase and no contextualization of polysemous words, and more importantly for our purposes, a narrative of interest cannot be specified *a priori* by a short description; it must instead be statistically prevalent enough in the corpus to emerge as a topic on its own. The third relies on supervised classifiers trained to detect a predefined narrative from labeled examples: for instance, (**coan2021computer**) build a two-level taxonomy of climate-change contrarian claims and train a classifier on tens of thousands of manually annotated paragraphs; + papier de Salomé Do et Étienne Ollion aumgented scientist. Unlike the first two families, such classifiers, built on contextual embeddings, can achieve fine-grained semantic resolution but only at the cost of substantial human labeling effort for every narrative of interest, and they do not generalize to a newly introduced narrative without retraining.

The qualitative leap we seek is the ability to measure narrative prevalence at a fine level of semantic nuance without either family's cost: neither reduced to a fixed vocabulary or a corpus-driven topic, nor requiring labeled examples per narrative. Just as importantly, we seek to compare the resulting measurements *across* narratives. This requires prevalence estimates for different narratives to sit on a common, mutually comparable scale, not merely to be individually well-behaved in time. The methodology we propose specifies a narrative *a posteriori* by a short descriptive text, and measures its prevalence directly against an unlabeled corpus, in the spirit of zero-shot, embedding-based text classification (Schopf, Braun, and Matthes 2023).

Challenges. Measuring relative narrative strength poses two distinct challenges.

1. **Idiosyncratic vs. common co-movements.** Even apparently opposed narratives share a common semantic direction (the *elections* axis) that causes their cosine scores to rise and fall in unison for reasons unrelated to their proper meaning. Disentangling idiosyncratic from common variation is therefore essential.
2. **Comparability across narratives.** Comparing the relative prevalence of two narratives requires their estimates to be measured on a common scale. With raw, non-orthogonal candidate vectors, this comparability fails on two counts: the measured signal for one narrative is contaminated by leakage from every other narrative sharing semantic content with it, and nothing ties the scale on which one narrative's presence is measured to that of another, since each is built from an independently chosen descriptive text.

Contributions. We propose an algebraic framework that simultaneously addresses both challenges. Our key contributions are:

- A linear generative model for article embeddings, articulated around the abstract, idealized concept of a *pure narrative*, a collection of orthonormal vectors that exists only in the latent embedding space, not as any literal piece of text. The model provides a rigorous statistical foundation for measuring narrative prevalence, yielding an unbiased estimator under pure narratives and formally exposing the correlation bias incurred when monitoring narrative prevalence via raw cosine similarity to non-orthogonal candidate texts (Section 3).

- Building on this diagnosis, a global orthogonalization procedure on the Stiefel manifold that constructs, from an arbitrary collection of candidate narrative texts, the corresponding collection of pure narratives: for each candidate, a representative vector that is close to the original text while being orthogonal to all others (Section 5).
- An empirical validation on a publicly available labeled news dataset, including a counterfactual study contrasting the naive and orthogonalized measures, demonstrating qualitative and quantitative agreement with known historical events (Section 6).

## 3. A Generative Model for Narrative Embeddings

### 3.1 Pure Narratives: an Idealized Abstraction

We take a *narrative* to be an abstract idea, claim, or framing about the world: an assertion, an interpretation of events, or a story that can be told about them. As a result, a piece of text can be a *representation* of a given narrative without being the narrative itself. Text is not the only possible representation, however. Language models make it possible to build a bridge between the qualitative world of sentences and words and the quantitative world. A narrative can then be represented as a vector in a latent space $\mathcal{H} \subset \mathbb{R}^d$, so that comparing or measuring narratives reduces to standard geometric operations on $\mathcal{H}$. This is a truly Platonist claim: just as Plato's Ideas live in their own world and language is only a support for them, *narratives* live in their own world, and texts or vectors are only supports for them.

However, by abuse of terminology, we will sometimes identify a narrative with its representation (either a text or a vector).

Collection of Pure Narratives Rather than studying a single narrative in isolation, this paper is concerned with studying relative prevalence among several narratives. We accordingly introduce the notion of a *collection of pure narratives*, whose representations live in the embedding space $\mathcal{H}$. A collection of pure narratives is a collection of narratives that are semantically independent from one another: each retains only an idiosyncratic component, separate from whatever is common across the collection. Purity in this sense is not a property of a single narrative taken in isolation, it is only meaningful relative to the collection it belongs to. In the embeddings space, this translate as followed:

**Definition 1** (Collection of pure narratives)**.** *A collection $V_1, \dots, V_K$ of narratives in $\mathcal{H} \subset \mathbb{R}^d$ is a collection of* pure narratives *if the collection $(V_i)_{1 \le i \le K}$ is orthonormal.*

Pure narratives should be understood as an idealized abstraction rather than a text we expect to observe or write down directly: we do not claim that any collection of texts is an exact representation of a collection of pure narratives, and we make no attempt to produce one. What matters at this stage is only that such an orthonormal collection *exists* in $\mathcal{H}$; Section 5 then shows how to recover, from any (impure) list of narrative-representing texts, a collection of vectors that is a good practical proxy for it.

### 3.2 A Generative Model on the Embedding Space

Given a family of pure narratives of interest $(V_i)_{1 \le i \le K}$, our goal is to study their relative prevalence within a corpus of texts. We introduce a probabilistic generative model for this purpose.

Setup. We fix, for each $i \le K$, a function $p_i : [0, \infty) \to [0, 1]$. We consider the random variables $(\alpha_1, ..., \alpha_K, \epsilon)$. We place ourselves on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ generated by the law of $(\alpha_1, \dots, \alpha_K, \varepsilon)$ such that: to an occurrence $\omega \in \Omega$ that occurs at time $\tau(\omega)$, we associate $\alpha_i(\omega) \in \{0, 1\}$ $(i \le K)$ and $\varepsilon(\omega) \in \mathcal{V}^\perp := \mathrm{Vect}(V_1, \dots, V_K)^\perp$, whose joint law, conditional on $\tau(\omega) = t$, is: the $\alpha_i(\omega)$ are Bernoulli with parameter $p_i(t)$, and $\varepsilon(\omega) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ on $\mathcal{V}^\perp$, independent of $(\alpha_i(\omega))_{i \le K}$.

Adding time via $\tau(\omega)$ equips $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $(\mathcal{F}_t)_{t\geq 0}$: given a corpus of occurrences $\omega_1, \omega_2, \ldots$, $\mathcal{F}_t$ records which occurrences have arisen by time $t$, together with the mark $(\alpha_i(\omega_n))_{i\leq K}, \varepsilon(\omega_n)$ of every occurrence $n$ with $\tau(\omega_n) < t$. By construction, $\tau$ is an $(\mathcal{F}_t)$-stopping time.

We introduce the random variable $X$ and assume that $X(\omega)$ decomposes as

$$X(\omega) = \sum_{i=1}^{K} \alpha_i(\omega)\, V_i + \varepsilon(\omega). \tag{1}$$

To keep the notation light, we drop the dependence on $\omega$: given a database of $N$ occurrences, we write $X_n := X(\omega_n)$, $\alpha_{i,n} := \alpha_i(\omega_n)$, $\varepsilon_n := \varepsilon(\omega_n)$, and $t_n := \tau(\omega_n)$, so that (1) reads

$$X_n = \sum_{i=1}^{K} \alpha_{i,n}\, V_i + \varepsilon_n, \qquad n = 1, \ldots, N. \tag{2}$$

Since $(V_i)_{i\leq K}$ are orthonormal, they span a subspace $\mathcal{V} := \mathrm{Vect}(V_1, \ldots, V_K)$, and $\mathcal{H} = \mathcal{V} \oplus \mathcal{V}^{\perp}$. Consequently, every $X_n \in \mathcal{H}$ admits a unique decomposition $X_n = X_n^{\parallel} + X_n^{\perp}$ with $X_n^{\parallel} \in \mathcal{V}$ and $X_n^{\perp} \in \mathcal{V}^{\perp}$, matching (2) with $\alpha_{i,n} = V_i^{\top} X_n$ and $\varepsilon_n = X_n^{\perp}$.

Qualitative interpretation. If one considers a corpus of texts (for instance a news corpus) with $N$ occurrences, the collection $(X_n)_{1\leq n\leq N}$ represents the corpus, more precisely, each $X_n$ is the vector embedding of a text (for instance of an article). The variable $\alpha_{i,n}$ indicates whether narrative $V_i$ is present in text $n$. The residual $\varepsilon_n$ captures everything in the text unrelated to the $K$ narratives of interest: for instance, author style, secondary facts, or unmodeled topics. As it lies in $\mathcal{V}^{\perp}$, it carries no component along any narrative of interest.

### *3.3 Temporal Conditioning and Prevalence*

Time discretization Time is discretized into periods indexed by $t \in \{1, \ldots, T\}$, corresponding to the values taken by the stopping time $\tau$ once the observation window is partitioned into days, weeks, or months, without fixing *a priori* the granularity. For each period $t$, we denote by

$$\mathcal{C}_t := \{n : \tau(\omega_n) = t\}, \qquad n_t := |\mathcal{C}_t| \tag{3}$$

the set and number of articles published during period $t$. Since $\tau$ is an $(\mathcal{F}_t)$-stopping time, $\{\tau(\omega_n) = t\}$ is $\mathcal{F}_t$-measurable for each $n$, and hence so are $\mathcal{C}_t$ and $n_t$.

Conditional prevalence. Recall from Section 3.2 that $p_i(t)$ denotes the Bernoulli parameter governing the presence of narrative $i$ at time $t$: $\mathbb{E}[\alpha_{i,n} \mid \mathcal{F}_t] = p_i(t)$ for any $n \in \mathcal{C}_t$. This quantity is not directly observable, since $\alpha_{i,n}$ is itself latent. Only the embedding $X_n$ is.

But, we leverage the properties of our model: the orthonormality of the $V_k$ and the fact that $\varepsilon_n \in \mathrm{Vect}(V_1, ..., V_K)^{\perp}$ to obtain:

$$\mathbb{E}\big[V_i^{\top} X_n \mid \mathcal{F}_t\big] = \sum_{k=1}^{K} \underbrace{\mathbb{E}\big[\alpha_{k,n} \mid \mathcal{F}_t\big]}_{=p_k(t)} \underbrace{V_i^{\top} V_k}_{=\delta_{ik}} + \mathbb{E}\big[\underbrace{V_i^{\top} \varepsilon_n}_{=0} \mid \mathcal{F}_t\big] = p_i(t), \tag{4}$$

This shows that $p_i(t)$ coincides with the conditional expectation of the observable projection $V_i^{\top} X_n$, which is what makes it estimable in practice.

**Proposition 1.**

$$\mathbb{E}\big[V_i^{\top} X_n \mid \mathcal{F}_t\big] = p_i(t). \tag{5}$$

Absolute frequency. The absolute frequency of narrative $i$ at time $t$ jointly captures narrative intensity and publication volume:

$$q_i(t) := \mathbb{E}\left[V_i^\top X_n \cdot \mathbf{1}_{\tau(\omega_n)=t}\right]. \tag{6}$$

Co-occurrence. For $i, j \leq K$, we define the co-occurrence of narratives $i$ and $j$ at time $t$ as

$$p_{ij}(t) := \mathbb{E}\left[\alpha_{i,n}\alpha_{j,n} \mid \mathcal{F}_t\right], \qquad n \in \mathcal{C}_t, \tag{7}$$

the conditional probability that an article published at $t$ carries both narratives $i$ and $j$ simultaneously; note $p_{ii}(t) = p_i(t)$. As with prevalence, $p_{ij}(t)$ is not directly observable, since $\alpha_{i,n}, \alpha_{j,n}$ are themselves latent. The following proposition shows it is recovered exactly from the observable product of projections $(V_i^\top X_n)(V_j^\top X_n)$.

**Proposition 2.** *For any $n \in \mathcal{C}_t$,*

$$(V_i^\top X_n)(V_j^\top X_n) = \alpha_{i,n}\alpha_{j,n} \quad a.s., \qquad \text{and hence} \qquad \mathbb{E}\left[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t\right] = p_{ij}(t). \tag{8}$$

*Proof.* Since $V_i, V_j \in \mathcal{V}$ and $\varepsilon_n \in \mathcal{V}^\perp$ identically, $V_i^\top \varepsilon_n = V_j^\top \varepsilon_n = 0$ almost surely. Hence, using (2) and $V_k^\top V_l = \delta_{kl}$, $V_i^\top X_n = \alpha_{i,n}$ and $V_j^\top X_n = \alpha_{j,n}$ exactly. The product identity follows immediately, and taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ recovers $p_{ij}(t)$ by definition. □

### 3.4 Empirical Estimators

This naturally leads to unbiased empirical estimators for the prevalence, the absolute frequency, and the co-occurrence.

$$\hat{p}_i(t) = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} V_i^\top X_n, \tag{9}$$

$$\hat{q}_i(t) = \frac{1}{N} \sum_{n \in \mathcal{C}_t} V_i^\top X_n = \frac{n_t}{N}\, \hat{p}_i(t). \tag{10}$$

$$\hat{p}_{ij}(t) = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} (V_i^\top X_n)(V_j^\top X_n). \tag{11}$$

**[JAMM: Peut-être qu'on pourra supprimer ces preuves qui sont quasi immédiates]**

Unbiasedness of $\hat{p}_i(t)$. Conditioning on $\mathcal{F}_t$:

$$\mathbb{E}\left[\hat{p}_i(t) \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} \mathbb{E}\left[V_i^\top X_n \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} p_i(t) = p_i(t). \tag{12}$$

By the tower property, $\hat{p}_i(t)$ is therefore an unbiased estimator of $p_i(t)$:

$$\mathbb{E}\left[\hat{p}_i(t)\right] = \mathbb{E}\Big[\mathbb{E}\left[\hat{p}_i(t) \mid \mathcal{F}_t\right]\Big] = p_i(t). \tag{13}$$

Unbiasedness of $\hat{q}_i(t)$. Because $N$ is fixed, we can exchange expectation and summation directly:

$$\mathbb{E}\left[\hat{q}_i(t)\right] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}\left[V_i^\top X_n \cdot \mathbf{1}_{\tau(\omega_n)=t}\right] = q_i(t). \tag{14}$$

Unbiasedness of $\hat{p}_{ij}(t)$. By Proposition 2, $(V_i^\top X_n)(V_j^\top X_n) = \alpha_{i,n}\alpha_{j,n}$ almost surely for every $n \in \mathcal{C}_t$, so conditioning on $\mathcal{F}_t$:

$$\mathbb{E}\big[\hat{p}_{ij}(t) \mid \mathcal{F}_t\big] = \frac{1}{n_t}\sum_{n\in\mathcal{C}_t}\mathbb{E}\big[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t\big] = \frac{1}{n_t}\sum_{n\in\mathcal{C}_t} p_{ij}(t) = p_{ij}(t), \tag{15}$$

and, by the tower property, $\hat{p}_{ij}(t)$ is an unbiased estimator of $p_{ij}(t)$:

$$\mathbb{E}\big[\hat{p}_{ij}(t)\big] = \mathbb{E}\Big[\mathbb{E}\big[\hat{p}_{ij}(t) \mid \mathcal{F}_t\big]\Big] = p_{ij}(t). \tag{16}$$

## 4. Failure of the Naive Approach: why orthonormality matters

At this stage, one could ask what's the point in having the orthonormality assumption, and having designed a generative model for text embeddings. Indeed, let's return to our baseline objective: monitoring the relative prevalence of a collection of narratives $(V_1, \dots, V_K)$ understood here in the abstract sense of Section 3.2.

One could claim that a simple way to proceed would be to write down a textual representation of each $V_i$ and pass it through a language model, obtaining candidate vectors $N_i \in \mathcal{H}$. Empirically, such candidates do not form an orthonormal family: normalizing them to unit norm is immediate, but nothing in how language models produce embeddings guarantees that they come out pairwise orthogonal.

The theoretical framework developed in Section 3 lets us establish precisely why this approach fails. We quantify it both at the level of prevalence (a first-moment quantity, Section 4.1) and at the level of co-occurrence, Section 4.2).

### *4.1 Prevalence*

**[TODO: Même question: garde-t-on les preuves ici ou en annexe ]**

Let $(N_i)_{1\le i\le K} \in \mathcal{H}$ be a family of unit-vector embeddings that represents our studied narratives. The family is not asssumed to be orthogonal. Analogously to Section 3.3, the naive approach measures the prevalence of $N_i$ at time $t$ as

$$\pi_{N_i}(t) := \mathbb{E}\big[N_i^\top X_n \mid \mathcal{F}_t\big], \qquad n \in \mathcal{C}_t. \tag{17}$$

Since $N_i$ and $X_n$ are both unit-norm, $N_i^\top X_n$ is exactly their cosine similarity; $\pi_{N_i}(t)$ is then the average cosine similarity between narrative $i$'s candidate embedding and the articles published during period $t$, precisely the quantity the naive approach would compute in practice.

Since $(V_k)_{k\le K}$ is an orthonormal basis of $\mathcal{V}$, $N_i$ decomposes uniquely as

$$N_i = \sum_{k=1}^{K} c_{ik}\, V_k + N_i^\perp, \qquad c_{ik} := N_i^\top V_k, \quad N_i^\perp \in \mathcal{V}^\perp. \tag{18}$$

**Proposition 3** (Naive prevalence under an impure candidate)**.**

$$\pi_{N_i}(t) = c_{ii}\, p_i(t) + \sum_{k\neq i} \underbrace{c_{ik}\, p_k(t)}_{\textit{leakage}}. \tag{19}$$

*Proof.* Using (18) and $X_n = \sum_l \alpha_{l,n} V_l + \varepsilon_n$,

$$N_i^\top X_n = \Big(\sum_k c_{ik} V_k + N_i^\perp\Big)^\top \Big(\sum_l \alpha_{l,n} V_l + \varepsilon_n\Big)$$
$$= \sum_k c_{ik}\, \alpha_{k,n} + (N_i^\perp)^\top \varepsilon_n,$$

using $V_k^\top V_l = \delta_{kl}$, $N_i^\perp \perp V_l$, and $V_k \perp \varepsilon_n$. Taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$, the term $\mathbb{E}[(N_i^\perp)^\top \varepsilon_n \mid \mathcal{F}_t] = 0$ since $\varepsilon_n$ is independent of $\mathcal{F}_t$ and centered, and $\mathbb{E}[\alpha_{k,n} \mid \mathcal{F}_t] = p_k(t)$, giving the result. □

Interpretation. When $(N_i)_i$ is pure, $c_{ii} = 1$ and $c_{ik} = 0$ for $k \neq i$, and $\pi_{N_i}(t) = p_i(t)$ exactly, recovering the unbiased estimator of Proposition 1. For an impure candidate, however, $\pi_{N_i}(t)$ mixes the true prevalence $p_i(t)$ with every other narrative's prevalence $p_k(t)$, weighted by the leakage coefficient $c_{ik} = N_i^\top V_k$. If a historical event drives up the prevalence of an unrelated but semantically connected narrative $k$ ($p_k(t)$ increases), the naive indicator $\pi_{N_i}(t)$ increases mechanically, creating the illusion that narrative $i$ itself has become more prevalent.

### 4.2 Co-occurrence of Two Narratives

Recall from Section 3.3 that the true co-occurrence of narratives $i$ and $j$ at time $t$, $p_{ij}(t) := \mathbb{E}[\alpha_{i,n}\alpha_{j,n} \mid \mathcal{F}_t]$, coincides exactly with $\mathbb{E}[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t]$ for pure narratives (Proposition 2). For arbitrary candidates $W_i, W_j \in \mathcal{H}$, the naive approach measures their co-occurrence by transplanting this same estimator to the candidates it actually has access to *i.e* language-model embeddings of a textual description of each narrative, rather than the (unobservable) pure narratives themselves. Concretely, for each article $n$, $X_n^\top W_i$ is the cosine similarity between the article and candidate $i$; their product is large only when the article aligns strongly with *both* candidates at once, and small otherwise, so that averaging this product over the pool of articles published at $t$ is, intuitively, a natural proxy for how often (and how strongly) narratives $i$ and $j$ jointly manifest in the corpus:

$$M_{ij}(t) := \mathbb{E}\big[(X_n^\top W_i)(X_n^\top W_j) \mid \mathcal{F}_t\big], \qquad n \in \mathcal{C}_t. \tag{20}$$

**Proposition 4** (Naive co-occurrence under impure candidates)**.** *Write* $W_i = \sum_k c_{ik} V_k + \eta_i$ *and* $W_j = \sum_l c_{jl} V_l + \eta_j$ *as in* (18)*. Then*

$$M_{ij}(t) = c_{ij}^2 p_{ij}(t) + \underbrace{\sum_{k\neq i,j} \sum_{l\neq i,j} c_{ik}\, c_{jl}\, p_{kl}(t)}_{\textit{signal leakage}} + \underbrace{\sigma_\varepsilon^2\, \eta_i^\top \eta_j}_{\textit{noise term}}. \tag{21}$$

*Proof.* As in the proof of Proposition 3, $X_n^\top W_i = \sum_k c_{ik}\alpha_{k,n} + \eta_i^\top \varepsilon_n$, and similarly for $W_j$. Expanding the product and taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$, the cross terms between $(\alpha_{k,n})_k$ and $\varepsilon_n$ vanish (independence, $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_t] = 0$), leaving

$$M_{ij}(t) = \sum_{k,l} c_{ik} c_{jl}\, \mathbb{E}[\alpha_{k,n}\alpha_{l,n} \mid \mathcal{F}_t] + \mathbb{E}\big[(\eta_i^\top \varepsilon_n)(\eta_j^\top \varepsilon_n) \mid \mathcal{F}_t\big].$$

The first sum is the signal term by definition of $p_{kl}(t)$; the second term equals $\sigma_\varepsilon^2\, \eta_i^\top \eta_j$ since $\eta_i, \eta_j \in \mathcal{V}^\perp$ and $\varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ on $\mathcal{V}^\perp$. □

Pure candidates. When $W_i = V_i$ and $W_j = V_j$, $c_{ik} = \delta_{ik}$, $c_{jl} = \delta_{jl}$, and $\eta_i = \eta_j = 0$: both the signal term collapses to $p_{ij}(t)$ and the noise term vanishes identically, so $M_{ij}(t) = p_{ij}(t)$ exactly. This is just what we had in Proposition 2.

Impure candidates When $W_i, W_j$ are not pure, (21) shows that $M_{ij}(t)$ departs from $p_{ij}(t)$ through two distinct mechanisms, each sufficient on its own to bias the naive measure.

*Signal leakage.* The signal term mixes *every* entry of the true co-occurrence matrix $(p_{kl}(t))_{k,l}$, not just $p_{ij}(t)$ itself. In particular, even if narratives $i$ and $j$ never truly co-occur ($p_{ij}(t) = 0$), $M_{ij}(t)$ can be large because $W_i$ and $W_j$ both carry a nonzero loading on a shared third narrative $k$ ($c_{ik} \neq 0$ and $c_{jk} \neq 0$): the naive approach then reports an artificial synchronization between two narratives that share nothing in the corpus.

*Semantic Noise* The noise term, $\sigma_\varepsilon^2\, \eta_i^\top \eta_j$, is an independent second source of artificial co-variation. Even a candidate pair with perfectly targeted signal ($c_{ik} = \delta_{ik}$, $c_{jl} = \delta_{jl}$, zero leakage onto any other narrative) still registers a spuriously elevated $M_{ij}(t)$ as soon as their residuals $\eta_i, \eta_j$ (whatever in each candidate's textual representation is unrelated to any narrative, e.g. shared vocabulary or phrasing used to describe both) happen to be non-orthogonal. Since $\sigma_\varepsilon^2 > 0$, any $\eta_i^\top \eta_j > 0$ mechanically inflates $M_{ij}(t)$ above the true co-occurrence, regardless of whether the narratives themselves ever co-occur in the corpus: two narratives can appear artificially correlated purely because their candidate descriptions happen to be phrased in a similar style.

The naive approach therefore fails on two distinct counts. At the level of prevalence, it conflates a narrative's true presence with leakage from every other narrative sharing representational overlap with it (Proposition 3). At the level of co-occurrence, it conflates genuine joint presence with two sources of spurious correlation: leakage onto narratives, and alignment of narrative-unrelated residuals, induced purely by the geometry of the candidate representations (Proposition 4). Both failure modes vanish exactly, once the candidates are replaced by their pure (orthogonalized) counterparts which is precisely what the procedure of Section 5 constructs.

## 5. Orthogonalization of Narratives

### 5.1 The Problem: Correlated Narrative Vectors

Let $\{N_i\}_{i=1}^n \subset \mathbb{R}^d$ denote the embedding vectors of $n$ narratives, obtained by passing their textual descriptions through a language model. A fundamental difficulty is that narratives written in the same language, sharing a journalistic style and treating related themes, yield representations that are highly correlated in the embedding space. In our experiments (Section 6), pairwise cosine similarities between raw narrative vectors lie in the range $[0.91, 0.96]$.

A direct consequence is that the time series of cosine similarities $\cos(N_i, X_t)$ (where $X_t$ is the embedding of an article published at time $t$) evolve almost synchronously across narratives, reflecting global topic volume rather than the relative dominance of specific narratives.

The goal is therefore to construct representative vectors $\{V_i\}_{i=1}^n$ that satisfy:

1. **Faithfulness**: each $V_i$ is as close as possible to $N_i^* = N_i/\|N_i\|$;
2. **Orthonormality**: $V_i^\top V_j = \delta_{ij}$.

### 5.2 Monotonous Orthogonalization and Its Limits

A natural candidate is the Gram–Schmidt algorithm. However, Gram–Schmidt is strongly order-dependent: the first vector is unchanged while subsequent vectors undergo increasingly large deformations, introducing an arbitrary bias. Moreover, it does not optimize any global fidelity criterion. Importantly, because the procedure is not incremental, any addition or removal of a

narrative requires recomputing the entire orthogonal basis to guarantee optimality and order-independence.

### 5.3 Global Procrustes Alignment and Its Limits

**[TODO: citer une source pour procruste ? c'est visiblement un résultat très classique]**

A natural alternative is the Orthogonal Procrustes solution, which finds the matrix $V$ that minimizes the global Frobenius norm $\|N^* - V\|_F^2$. While this admits an exact closed-form solution via Singular Value Decomposition (SVD), it maximizes the *sum* of alignments rather than the *minimum*. In the context of highly correlated narratives, Procrustes optimization can sacrifice the representation of minority narratives: it allows individual vectors $V_i$ to be pushed far from $N_i^*$ if doing so slightly improves the overall global alignment. To prevent any single narrative from being discarded or severely distorted, we instead require a max-min formulation.

### 5.4 Global Optimization on the Stiefel Manifold

To overcome these limitations, we seek the orthonormal family $\{V_i\}_{i=1}^n$ that maximizes the minimum alignment with the original narratives, i.e. we solve the max-min problem:

$$\arg\max_{(V_1,\dots,V_n)} \min_{i=1,\dots,n} N_i^{*\top} V_i \qquad \text{subject to} \quad \begin{cases} \|V_i\| = 1 & \forall\, i, \\ N_i^{*\top} V_i \geq 0 & \forall\, i, \\ V_i^\top V_j = 0 & \forall\, i \neq j. \end{cases} \tag{22}$$

The constraint $N_i^{*\top} V_i \geq 0$ ensures that $V_i$ and $N_i^*$ are on the same side of the hyperplane, i.e. that $V_i$ is a genuine representative of $N_i$ rather than its antipode. The max-min objective guarantees that the algorithm focuses on the worst-case narrative, preventing any single narrative from being sacrificed.

Stacking the row vectors $V_i$ into a matrix $V \in \mathbb{R}^{n\times d}$, the orthonormality constraints are equivalent to $VV^\top = I_n$, i.e. $V$ belongs to the *Stiefel manifold* (Boumal 2023; Wen and Yin 2013):

$$\mathrm{St}(n,d) = \left\{ V \in \mathbb{R}^{n\times d} : VV^\top = I_n \right\}. \tag{23}$$

**[TODO: Doit-on garder le détail de la résolution du pb d'optimisation dans le texte ou le mettre dans l'annexe.]**

Because the objective $\min_i(N_i^{*\top} V_i)$ is non-differentiable, we approximate it with the smooth *softmin* parameterized by temperature $\tau > 0$:

$$f(V) = -\tau \log \sum_{i=1}^{n} \exp\left( -\frac{N_i^{*\top} V_i}{\tau} \right). \tag{24}$$

Minimizing $f$ on $\mathrm{St}(n,d)$ is performed via *Riemannian gradient descent*, with three steps per iteration:

1. **Euclidean gradient.**

$$\nabla f(V) = -\operatorname{diag}(w)\, N, \qquad w_i = \frac{\exp(-N_i^{*\top} V_i/\tau)}{\sum_j \exp(-N_j^{*\top} V_j/\tau)}. \tag{25}$$

2. **Riemannian gradient.** Project onto the tangent space of $\mathrm{St}(n,d)$ at $V$:

$$\operatorname{grad} f(V) = \nabla f(V) - V \operatorname{sym}\big(V^\top \nabla f(V)\big), \qquad \operatorname{sym}(A) = \tfrac{1}{2}(A + A^\top). \tag{26}$$

3. **Retraction.** After the descent step $\tilde{V} = V - \alpha \operatorname{grad} f(V)$, the updated matrix $\tilde{V}$ no longer necessarily satisfies the orthonormality constraint defining the Stiefel manifold (i.e., $\tilde{V}^\top \tilde{V} \neq I$ in general).
   To map $\tilde{V}$ back onto the Stiefel manifold $\mathrm{St}(p, n)$, we apply a retraction based on the QR decomposition. Concretely, we compute the QR factorization $\tilde{V} = QR$, where $Q$ has orthonormal columns and $R$ is upper triangular. The retracted point is then defined as $V_{\text{new}} = Q$.
   As discussed in Absil et al. Absil, Mahony, and Sepulchre 2009, Chapter 4, standard QR-based retractions are used to project iterates back onto the Stiefel manifold after a descent step.

This iterative scheme guarantees that the orthonormality constraints are satisfied at every iteration while converging to a local optimum of (22).

### 5.5 Interpretation of the Orthogonalized Vectors

We test our procedure with narratives that can be found in Appendix 1. The four panels of Figure 1 and 2 illustrate the behavior of the original narrative embeddings and their orthogonalized counterparts under both the Stiefel-based procedure and the classical Gram-Schmidt algorithm.

Panel (a) displays the cosine similarity matrix between the raw narrative embeddings. A striking feature is the overall high level of correlation across all narrative pairs, reflecting the anisotropy of the embedding space produced by *ModernBERT*. Diagonal blocks are systematically darker, confirming that narratives belonging to the same thematic chapter are more strongly correlated with one another than with narratives from other chapters. For instance, *"AI Disruption of Labor Markets"* and *"Big Tech Race and Regulatory Pushback"* exhibit a cosine similarity of 0.95, whereas the similarity between *"AI Disruption of Labor Markets"* and *"Financial Market Volatility and Fed Signaling"* is lower (with cosine similarity of 0.91). This intra-chapter cohesion validates the semantic consistency of the narrative groupings.

Panel (b) demonstrates the success of the Stiefel optimization. The resulting representative matrix is fully orthogonal, with pairwise cosine similarities equal to zero across all narrative pairs. Crucially, this orthogonality holds not only across thematic chapters but also within chapters: *"Extreme Heat and Emergency Policy Response"* is equally orthogonal to *"International Climate Commitments Under Pressure"* and *"US China Rivalry and Indo Pacific"*. The procedure thus isolates the idiosyncratic component of each narrative, stripping away both cross-chapter and intra-chapter shared variance. What remains is the semantic signal that is strictly proper to each narrative, independently of all others.

Panel (c) presents the cross-similarity between raw narratives and their Stiefel-optimized representatives. A clear diagonal structure emerges: each representative is more strongly correlated with its own source narrative than with any other, confirming that the optimization procedure preserves the idiosyncratic semantic content of each narrative while beginning to disentangle inter-narrative correlations.

Panel (d) exposes the limitations of the classical Gram-Schmidt orthogonalization. This procedure is well known to be sensitive to the ordering of input vectors: by construction, the first vector in the sequence is left entirely unchanged, while subsequent vectors are progressively deflated. As a result, the first representative, here that of *"Extreme Heat and Emergency Policy Response"*, retains its original embedding and remains correlated with all other raw narratives. The pathological consequence is particularly striking: the representative of *"Extreme Heat and Emergency Policy Response"* exhibits a higher cosine similarity with the raw narrative *"Middle East Conflict and Humanitarian Crisis"*, for instance, than the representative of *"Middle East Conflict and Humanitarian Crisis"* does with the very narrative it is supposed to represent. This severe distortion renders the Gram-Schmidt basis semantically uninterpretable as a set of narrative representatives, and underscores the superiority of the Stiefel-based approach for this task.

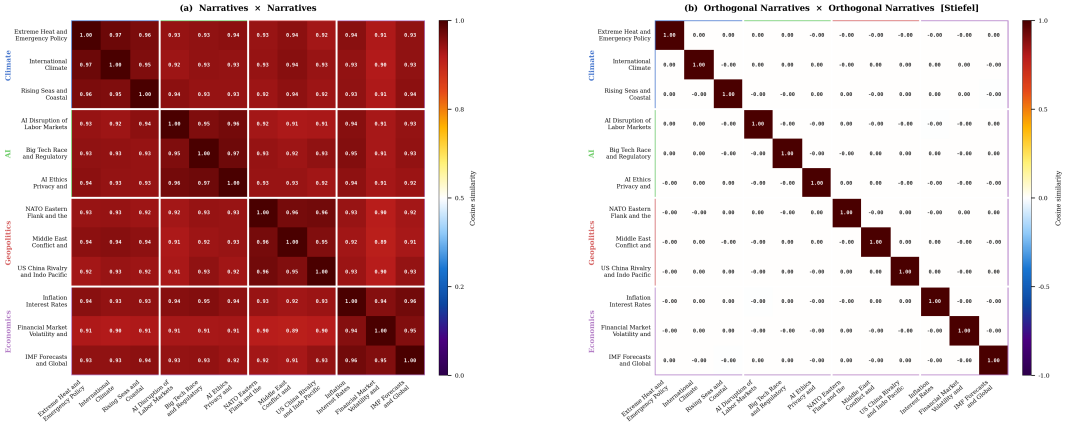



**Figure 1.** Cosine similarity matrices of narrative embeddings before and after Stiefel orthogonalization. Panel (a) shows the raw *ModernBERT* embeddings, where diagonal blocks are systematically darker, reflecting stronger intra-chapter semantic cohesion. Panel (b) shows the orthogonalized embeddings after Stiefel optimization, yielding zero pairwise cosine similarity across all narrative pairs thereby isolating the idiosyncratic semantic signal of each narrative.

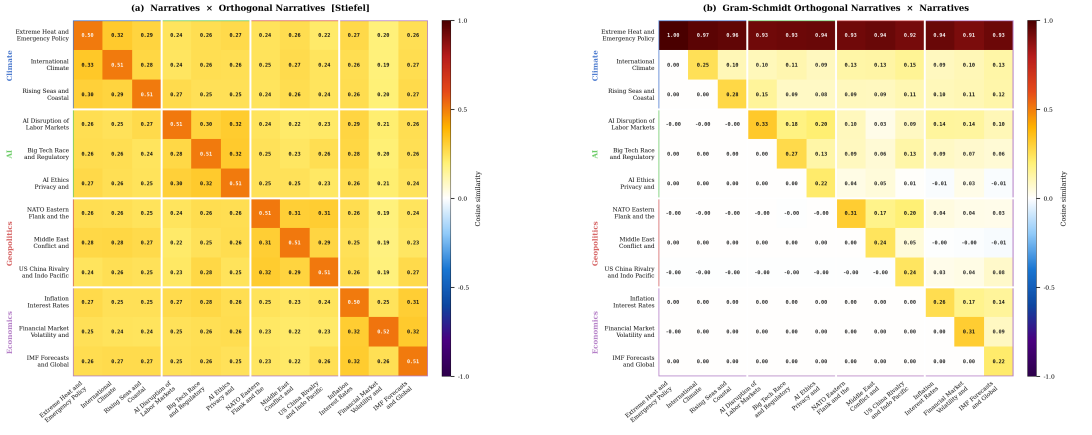



**Figure 2.** Cross-similarity matrices between raw narrative embeddings and their orthogonalized representatives. Panel (c) shows the Stiefel-optimized representatives, where a clear diagonal structure confirms that each representative retains the idiosyncratic semantic content of its source narrative. Panel (d) shows the Gram-Schmidt representatives, revealing a severe order-dependent distortion: the first narrative is left unchanged and remains correlated with all others, to the point where its representative exhibits higher similarity with unrelated raw narratives than some narratives' own representatives do with themselves.

## 6. Experiments and Validation on a Dataset

[CAL: expliquer qu'on utilise cette base car elle va permettre le contrefactuel + mettre en place le framework probabiliste pour les tests] [CAL: alternatives à modernBERT] [JAMM: première recherche: https://chatgpt.com/share/69e58d30-f2c4-8329-aa4f-5682054e5775]

The objective of this empirical study is to validate the methodology introduced in the previous sections from two complementary perspectives. More precisely we will compare those two approaches:

- We use the embeddings that directly encodes the text representations of the studied collection of

narratives. We denote them as $(N_i)_i$ and we will sometimes refer to them as "raw narratives ".
- We use the vectors obtained after applying the procedure described in 5.4. We denote them as $(V_i)_i$ and we will sometimes refer to them as "pure narratives "or "orthogonalized narratives ".

We want to check that:

- That orthogonalized narratives effectively remove the biases highlighted in 2;
- The orthogonalized representatives preserve the meaning;
- Tracking a document's similarity to an orthogonalized narrative is a more meaningful way to monitor the exposure to that specific narrative (compared to using the raw narratives)

To this end, we rely on the *News Category Dataset* (Misra 2022), whose manually assigned categories provide a natural counterfactual framework.

### *6.1 Dataset and Experimental Setup*

Dataset. We use the *News Category Dataset* (Misra 2022), which contains approximately 210 000 headlines published by *HuffPost* between 2012 and 2022. Since the archive becomes sparse after May 2018, we restrict the analysis to the 2012–2018 period, resulting in approximately 200 000 articles.

Among all available categories, we retain four sufficiently represented chapters: *Politics* (32 739 articles), *Entertainment* (16 058), *Travel* (9 887), and *Parenting* (8 677 articles). These four chapters naturally match the narrative groups introduced below and provide labeled subsets that can be exploited for validation.

Temporal resampling. The raw dataset is highly unbalanced over time, particularly for the Politics category during the 2016 presidential campaign. To avoid interpreting variations in publication volume as changes in narrative prevalence, we perform a temporal resampling that enforces a constant weekly number of articles while preserving the relative proportions across categories.Table 1 reports the resulting distribution.

**[TODO: Ici, il y a sans doute un commentaire à faire sur les catégories (les seules ok à mon avis c'est entertainment et politics) - ce qui n'empêche pas d'utiliser les autres si on moyenne sur toute la période]**

**Table 1.** Number of articles per category and year after resampling.

| Year | Entertainment | Parenting | Politics | Travel |
|---|---|---|---|---|
| 2012 | 792 | 4,087 | 0 | 4,221 |
| 2013 | 1,154 | 4,710 | 0 | 3,786 |
| 2014 | 2,445 | 1,403 | 3,684 | 2,079 |
| 2015 | 3,614 | 0 | 5,351 | 705 |
| 2016 | 2,478 | 0 | 6,885 | 311 |
| 2017 | 2,194 | 0 | 7,273 | 178 |
| 2018 | 1,375 | 0 | 2,450 | 60 |

Embeddings. Both news headlines and narrative descriptions are embedded in the same 768-dimensional semantic space using *ModernBERT* (Warner, Mishra, Stolfo, et al. 2024). Narrative descriptions were generated with the assistance of a large language model to ensure a homogeneous writing style.

### 6.2 Narratives

For each of the four chapters, we manually define three representative narratives intended to capture the main social, political, and cultural themes that characterize the corresponding articles over the 2012–2018 period. The complete list of the twelve narratives is provided in Appendix 2. A selection is shown below.

**Table 2.** Narratives used in the analysis (selection).

| Chapter | Narrative | Description |
|---|---|---|
| Politics | Trump as Disruptor and Institutional Stress | Donald Trump rises to power by channeling distrust toward political elites, media institutions, and bureaucratic authority, presenting himself as a disruptive outsider challenging democratic norms. |
| Entertainment | Entertainment as Political Resistance | Film, television, comedy, and celebrity culture become arenas of political contestation blending entertainment with explicit political messaging. |
| Parenting | The Overwhelmed Modern Parent | Parenthood is portrayed as emotionally intense and structurally unsupported, with parents struggling to balance careers, childcare costs, and social expectations. |
| Travel | Experiential and Transformational Travel | Travel is framed as a path to personal growth and cultural immersion through encounters with landscapes, heritage sites, and local communities. |

The raw narrative embeddings are then orthogonalized using the Stiefel optimization procedure described in Section 5, producing the corresponding *collection of pure narratives* $(V_i)_i$.

### 6.3 First qualitative results

We first perform a qualitative comparison between the prevalence estimators obtained using the raw narrative vectors $N_i$ and their orthogonalized representatives $V_i$.

To facilitate visual interpretation, we restrict this analysis to the Politics and Entertainment chapters over the 2014–2018 period, where both categories contain a sufficiently balanced number of observations. The displayed curves correspond to an exponential moving average of the estimator $\hat{p}_i(t)$ introduced in Section 3.

The contrast between the two approaches is immediate. Under the raw narrative vectors, the prevalence series evolve almost perfectly synchronously, illustrating the leakage phenomenon anticipated in Section 3: when the prevalence of an unrelated but semantically connected narrative rises, the naive prevalence indicator computed on raw narratives increases mechanically along with it. Pure narratives, by contrast, produce clearly differentiated temporal trajectories, with each of them now exhibiting its own distinctive dynamics.

The clearest visual illustration of this contrast occurs between January 2017 and January 2018, when a pronounced bump appears in the raw prevalence series of virtually every narrative. Under the purified representation, this bump is observed almost exclusively in the *Welfare State Under Pressure* narrative, the only one for which it plausibly reflects genuine content rather than a shift shared indiscriminately across the corpus. This single episode captures the practical interest of orthogonalization: without it, an apparent surge in any given narrative could just as easily be an artifact of another, unrelated narrative's rise. This pattern is consistent with the historical record, as the early months of the Trump administration were marked by sustained legislative pressure on the social safety, most notably through repeated attempts to repeal the Affordable Care Act (Obamacare).




**Figure 3.** Time series of EMA weekly means of cosine similarities for raw and orthogonal narratives.

The observed behavior is also consistent with the historical context used to design the narratives. The political narratives were deliberately chosen to capture the rise of Donald Trump's presidential campaign and its associated political themes. Accordingly, the prevalence of the narratives *Trump as Disruptor and Institutional Stress* and *Culture War and Identity Conflict* increases steadily throughout the campaign and reaches its highest levels around the 2016 election. This evolution is qualitatively consistent with the growing importance of these themes in the political debate.

Conversely, the Entertainment narratives were intentionally designed to represent long-term structural trends rather than isolated historical events. Their estimated prevalence therefore remains comparatively stable over time, exhibiting only moderate fluctuations. The resulting time series are thus coherent with the expected temporal behavior of the corresponding narratives.

### *6.4 Co-variation, semantic specificity and rupture tests*

**[JAMM: TO DO: changer le titre]**

We next evaluate the proposed methodology through three complementary quantitative experiments.

Reduction of artificially high co-variation We first observe qualitatively that the prevalence curves within the political and Entertainment subsets of narratives evolve almost synchronously. We now turn to quantifying this phenomenon across the full set of narratives under study.

We use the metric introduced in Section 3:

$$M_{ij}(t) := \mathbb{E}\left[(X_n^\top W_i)(X_n^\top W_j) \mid \mathcal{F}_t\right], \qquad n \in \mathcal{C}_t.$$

As discussed there, for a collection of raw, non-orthogonalized narratives, two distinct effects artificially inflate this covariation: signal leakage and the correlation induced by the noise term. We therefore assess whether orthogonalization produces a sizable reduction in covariation. Time is discretized weekly: for each pair $(i, j)$ we compute $M_{ij}(t)$ and average it over the entire study period.

**Table 3.** Descriptive statistics of pairwise narrative covariation $M_{ij}(t)$ ($\times 10^{-7}$), before (N) and after (V) orthogonalization.

| | N (raw) | V (purified) |
|---|---|---|
| mean | 74.018 | 6.486 |
| std | 0.646 | 0.737 |
| median | 73.985 | 6.421 |
| min | 72.775 | 5.209 |
| max | 75.546 | 8.343 |

The empirical results strongly support this prediction: as reported in Table 3, pairwise covariation scores computed from the raw narrative vectors are, on average, approximately one order of magnitude larger than those obtained after orthogonalization.

Semantic specificity. The second validation exercise aims to verify that purified representatives preserve the semantic meaning of the original narratives.

The labeled structure of the News Category Dataset naturally provides a counterfactual framework: since every narrative belongs to one of four chapters, we compare its average cosine similarity within its own chapter to its average similarity outside that chapter.

For the orthogonalized representatives, we define

$$\Delta^{i,k}(V_{i,k}) = \mathbb{E}\left[(X^i)^\top V_{i,k}\right] - \mathbb{E}\left[(X^{\neg i})^\top V_{i,k}\right],$$

where $X^i$ denotes articles belonging to chapter $i$, while $X^{\neg i}$ denotes all remaining articles.

The analogous quantity is computed for the original narrative vectors,

$$\Delta^{i,k}(N_{i,k}) = \mathbb{E}\left[(X^i)^\top N_{i,k}\right] - \mathbb{E}\left[(X^{\neg i})^\top N_{i,k}\right].$$

A meaningful narrative should be more strongly represented within its own chapter than outside it, implying

$$\Delta^{i,k}(N_{i,k}) > 0, \qquad \Delta^{i,k}(V_{i,k}) > 0.$$

More importantly, if orthogonalization successfully removes semantic information shared with other chapters while preserving each narrative's intended meaning, we expect

$$\Delta^{i,k}(V_{i,k}) > \Delta^{i,k}(N_{i,k}).$$

The results are satisfactory: every chapter exhibits $\Delta^i(V_i) > 0$, whereas the Entertainment chapter and two narratives of Politics chapter have $\Delta^i(N_i) < 0$. Regarding $\Delta_{\text{gain}}$, it is positive for three chapters out of four.

**Table 4.** Sensitivity analysis: 1000 bootstrap draws, 100% of articles, 95% CI for $\Delta^i(V_i)$,$\Delta^i(N_i)$, and $\Delta_{\text{gain}} = \Delta^i(V_i) - \Delta^i(N_i)$

| Chapter | Narrative | $\Delta^i(N_i)$ | $\Delta^i(V_i)$ | $\Delta_{\text{gain}}$ |
|---|---|---|---|---|
| Politics | Trump as disruptor and institutional stress | 0.0019 [0.0015, 0.0023] | 0.0377 [0.0374, 0.0381] | 0.0358 [0.0353, 0.0363] |
| Politics | Culture war and identity conflict | −0.0002 [-0.0006, 0.0003] | 0.0312 [0.0309, 0.0316] | 0.0314 [0.0309, 0.0319] |
| Politics | Welfare state under press | −0.0053 [-0.0058, -0.0048] | 0.0084 [0.0078, 0.0090] | 0.0137 [0.0134, 0.0141] |
| Entertainment | Entertainment as political resistance | −0.0145 [-0.0151, -0.0139] | 0.0062 [0.0058, 0.0066] | 0.0207 [0.0203, 0.0210] |
| Entertainment | Franchise nostalgia and serial dominance | −0.0107 [-0.0112, -0.0101] | 0.0207 [0.0202, 0.0211] | 0.0313 [0.0309, 0.0318] |
| Entertainment | Celebrity image management in the social media... | −0.0153 [-0.0159, -0.0148] | 0.0054 [0.0050, 0.0058] | 0.0207 [0.0202, 0.0212] |
| Parenting | The overwhelmed modern parent | 0.0253 [0.0249, 0.0256] | 0.0118 [0.0114, 0.0123] | −0.0134 [-0.0139, -0.0129] |
| Parenting | Intensive and expert guided parenting | 0.0287 [0.0283, 0.0291] | 0.0302 [0.0297, 0.0308] | 0.0015 [0.0010, 0.0020] |
| Parenting | Raising children in a politicized society | 0.0244 [0.0240, 0.0248] | 0.0174 [0.0171, 0.0178] | −0.0070 [-0.0073, -0.0066] |
| Travel | Experiential and transformational travel | 0.0251 [0.0246, 0.0257] | 0.0298 [0.0293, 0.0304] | 0.0047 [0.0042, 0.0052] |
| Travel | Travel optimization and consumer strategy | 0.0279 [0.0274, 0.0285] | 0.0415 [0.0410, 0.0420] | 0.0135 [0.0130, 0.0141] |
| Travel | Aesthetic and instagrammable destinations | 0.0297 [0.0293, 0.0303] | 0.0452 [0.0446, 0.0458] | 0.0155 [0.0150, 0.0160] |

Narrative specificity of detected ruptures We now quantify the qualitative observation made above: ruptures detected on raw narrative representations tend to be shared across many, otherwise unrelated, narratives, whereas ruptures detected on purified representations are largely specific to a single narrative.

For a narrative representation $M \in \{N_i, V_i\}$ and each candidate date $t$, let $\mathcal{B}_t$ and $\mathcal{A}_t$ denote the sets of articles published, respectively, in the 90-day window preceding and following $t$. Writing $Q_n(M) = X_n^\top M$ for the projected score of article $n$, we test, at every candidate date $t$,

$$H_0 : \mathbb{E}\big[Q_n(M) \mid n \in \mathcal{B}_t\big] = \mathbb{E}\big[Q_n(M) \mid n \in \mathcal{A}_t\big],$$

using a two-sided Welch $t$-test, which does not require equal variances across the two windows. Because this test is repeated over hundreds of candidate dates, the resulting $p$-values are corrected for multiple testing using the Benjamini–Hochberg FDR procedure; consecutive significant dates are then grouped into a single detected rupture. This procedure is applied separately to the raw ($N$) and purified ($V$) representations of every narrative in our collection.

Results are reported in Figures **??** Three observations stand out.

First, the number of detected ruptures is systematically higher under $N$ than under $V$: averaged across narratives, 27.1 ruptures are detected under $N$, against 16.5 under $V$, and this ordering holds narrative by narrative without exception.

Second, pairwise overlap between the sets of rupture dates detected across different narratives is markedly higher under $N$ than under $V$, as measured by the Jaccard index[1]: the average pairwise Jaccard index is 0.74 under $N$, compared to only 0.25 under $V$. Taken together, these two findings confirm that raw representations tend to co-detect the same, corpus-wide ruptures, consistent with the leakage phenomenon described in Section **??**, whereas purified representations isolate ruptures that are substantially more specific to each individual narrative.

Third, this is directly illustrated by the July 2017 bump discussed in our qualitative analysis: under $N$, it is detected as a significant rupture for every narrative in the collection, while under $V$ it is detected almost exclusively for the *Welfare State Under Pressure* narrative, the only one for which it plausibly reflects genuine, narrative-specific content (as the period corresponds to Trump's first semester as President with notably repeated attempt to repeal the Obamacare) rather than a shift shared indiscriminately across the corpus.

1. Jaccard index: $J(A, B) = \dfrac{|A \cap B|}{|A \cup B|}$.

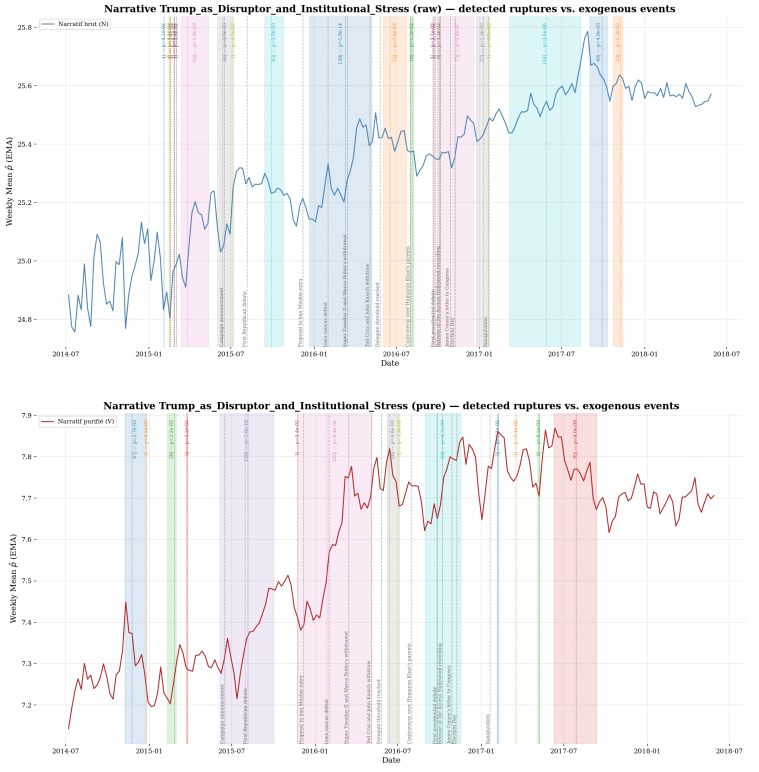



**Figure 4.** Time series of EMA weekly means of cosine similarities for raw and orthogonal Trump narrative, with detected ruptures as well as a set of exogenous events related to Trump's campaign.

## 7. Conclusion

We have introduced a rigorous algebraic methodology for monitoring the relative prevalence of competing narratives in a time-evolving text corpus. The key insight is that standard cosine similarity between article embeddings and raw narrative vectors suffers from a structural correlation bias induced by shared semantic content, which we characterize formally through a linear generative model.

Our remedy—a global orthogonalization on the Stiefel manifold—eliminates this shared component while preserving each narrative's idiosyncratic direction, yielding an unbiased, non-parametric, and interpretable prevalence estimator. Validation on a large labeled news corpus confirms that the estimator faithfully recovers known historical dynamics, whereas the naive approach fails.

Perspectives. The natural extension of this work is the application of the estimator to large-scale financial news corpora, opening the way to a direct quantitative model of narrative competition and its influence on the formation of economic expectations—a question at the intersection of computational linguistics and market microstructure.

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Funding Statement **[TODO: Insert funding statement or 'None'.]**

Competing Interests None.

Data Availability Statement The News Category Dataset is publicly available (Misra 2022). Replication code can be found at **[TODO: insert repository URL]**.

Ethical Standards The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions Conceptualization: J.A., M.M., C.-A.L. Methodology: J.A., M.M., C.-A.L. Data curation: J.A., M.M. Data visualisation: J.A., M.M. Writing original draft: J.A., M.M., C.-A.L. All authors approved the final submitted draft.

## Notes

**[TODO: Est-ce que c'est léger en termes de références ?]**

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## Appendix 1. Complete List of Narratives as an Illustration

**Table 5.** All narratives used in the analysis.

| Chapter | Narrative | Description |
|---|---|---|
| Climate | Extreme Heat and Emergency Policy Response | Record-breaking temperatures across southern Europe this summer triggered unprecedented heatwaves, forcing governments to activate emergency climate plans and accelerate the transition toward renewable energy sources in order to meaningfully reduce carbon emissions and limit further global warming. |
| Climate | International Climate Commitments Under Pressure | As climate-related disasters multiply worldwide, the United Nations urgently calls for stronger adherence to the Paris Agreement, emphasizing that industrialized nations must deliver substantial and verifiable reductions in greenhouse gas emissions before irreversible climate tipping points are reached. |
| Climate | Rising Seas and Coastal Vulnerability | The accelerating melting of Arctic glaciers, driven by sustained global warming, is causing a gradual but measurable rise in sea levels that threatens tens of millions of people living in low-lying coastal zones, according to a major scientific report released this week. |
| Artificial Intelligence | AI Disruption of Labor Markets | The rapid rise of generative artificial intelligence models is profoundly reshaping global labor markets, prompting technology companies to invest billions of dollars in autonomous systems capable of gradually replacing skilled workers across a wide range of industries and professional sectors. |
| Artificial Intelligence | Big Tech Race and Regulatory Pushback | Silicon Valley's leading technology firms are intensifying their competition in artificial intelligence by unveiling increasingly powerful language models, while European regulators work to constrain these developments through strict legislative frameworks designed to protect citizens' fundamental rights and digital freedoms. |
| Artificial Intelligence | AI Ethics, Privacy, and Public Sector Governance | Recent breakthroughs in artificial intelligence are raising critical ethical concerns around personal data privacy and cybersecurity, prompting several national governments to establish dedicated regulatory frameworks governing the deployment of these technologies within public institutions and essential services. |
| Geopolitics | NATO Eastern Flank and the Russia–Ukraine War | The ongoing military conflict between Russia and Ukraine continues to destabilize the European geopolitical order, compelling NATO allies to reinforce their eastern flank with additional troops and to significantly increase national defense budgets in response to sustained and escalating Russian aggression. |
| Geopolitics | Middle East Conflict and Humanitarian Crisis | The armed conflict in the Middle East has triggered a humanitarian catastrophe of alarming proportions, prompting urgent international calls for an immediate ceasefire and the rapid deployment of emergency aid to civilian populations trapped in active combat zones across the region. |
| Geopolitics | US–China Rivalry and Indo-Pacific Tensions | Growing geopolitical rivalry between Washington and Beijing over the status of Taiwan is seriously threatening stability across the Indo-Pacific region, with both superpowers escalating military demonstrations and economic sanctions in what increasingly resembles a renewed cold war confrontation. |

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Table 5 – *Continued from previous page*

| Chapter | Narrative | Description |
|---|---|---|
| Economics | Inflation, Interest Rates, and Household Strain | Persistent inflation across the world's major economies is forcing central banks to maintain elevated interest rates, placing significant pressure on economic growth and disproportionately burdening lower-income households struggling to absorb the rising cost of everyday goods and essential services. |
| Economics | Financial Market Volatility and Fed Signaling | Global financial markets experienced sharp volatility this week as investors reacted to the latest monetary policy signals from the United States Federal Reserve, triggering significant fluctuations across major stock indices and a notable depreciation of several emerging market currencies. |
| Economics | IMF Forecasts and Global Growth Slowdown | Global economic growth is decelerating at a concerning pace according to the latest International Monetary Fund projections, which identify trade tensions, rising public debt levels, and slowing emerging market performance as the primary drivers behind deteriorating worldwide economic prospects. |

## Appendix 2. Complete List of Narratives for the Analysis

**Table 6.** All narratives used in the analysis.

| Chapter | Narrative | Description |
|---|---|---|
| Politics | Trump as Disruptor and Institutional Stress | Donald Trump rises to power by channeling distrust toward political elites, media institutions, and bureaucratic authority, presenting himself as a disruptive outsider challenging democratic norms and reframing institutional resistance as partisan sabotage. |
| Politics | Culture War and Identity Conflict | American politics becomes increasingly structured around culture wars in which abortion, LGBTQ rights, immigration, race, religion, and gun control become symbolic battlegrounds mobilizing voters through moralized political conflict. |
| Politics | Welfare State Under Press | Political debates around health care reform, social assistance, and regulatory protections reflect a broader struggle over the future of the welfare state amid economic insecurity and ideological polarization. |
| Entertainment | Entertainment as Political Resistance | Film, television, comedy, and celebrity culture become arenas of political contestation where cultural production increasingly blends entertainment with explicit political messaging. |
| Entertainment | Franchise Nostalgia and Serial Dominance | The entertainment industry relies heavily on sequels, reboots, and cinematic universes, capitalizing on nostalgia and established IP in an increasingly competitive streaming environment. |
| Entertainment | Celebrity Image Management in the Social Media Age | Celebrities navigate a hyper-visible media ecosystem where personal relationships, identity declarations, and scandals shape public image and reputation management. |
| Parenting | The Overwhelmed Modern Parent | Parenthood is portrayed as emotionally intense and structurally unsupported, with parents struggling to balance careers, childcare costs, and social expectations. |
| Parenting | Intensive and Expert-Guided Parenting | Parenting becomes increasingly expert-driven, with reliance on psychological advice, scientific studies, and educational technologies to optimize children's development. |
| Parenting | Raising Children in a Politicized Society | Parents confront a social environment shaped by political polarization, gun violence debates, and shifting gender norms that influence children's values and civic identity. |
| Travel | Experiential and Transformational Travel | Travel is framed as a path to personal growth and cultural immersion through encounters with landscapes, heritage sites, and local communities. |
| Travel | Travel Optimization and Consumer Strategy | Travel is increasingly treated as a strategic consumer activity involving price optimization, loyalty programs, and efficiency strategies. |
| Travel | Aesthetic and Instagrammable Destinations | Destinations gain visibility through their visual appeal and shareability on social media platforms, emphasizing photogenic landscapes and unique experiences. |